Rotation conventions and closed flag walks

Human Compression project

The survey of Schulte–Weiss uses consecutive rotations, whereas our chamber arguments use rotations based at one fixed type. We give the change of generators and recover the survey's period formulas for holes and zigzags. The formulas are established results recorded in the survey; no new classification or answer to its Problem 7 is asserted.

1 Two choices of rotations

Let $n \geq 3$ and let a group G have generators $\sigma_1, \ldots, \sigma_{n-1}$ satisfying

$$(\sigma_i \sigma_{i+1} \cdots \sigma_j)^2 = 1 \qquad (1 \leq i < j \leq n-1).$$

Put $\alpha_0 = 1$ and $\alpha_i = \sigma_1 \cdots \sigma_i$. Then $\sigma_i = \alpha_{i-1}^{-1}\alpha_i$, and $\alpha_i^2 = 1$ for $i \geq 2$. These are identities in the marked group, independent of an assertion that its coset geometry is a polytope.

Proposition 1. *An automorphism θ simultaneously inverts every α_i if and only if*

$$\theta(\sigma_1) = \sigma_1^{-1}, \qquad \theta(\sigma_2) = \sigma_1^2\sigma_2, \qquad \theta(\sigma_i) = \sigma_i \quad (i \geq 3).$$

If such an automorphism exists, its square is the identity.

Proof. Suppose first that $\theta(\alpha_i) = \alpha_i^{-1}$. The assertion for $\sigma_1 = \alpha_1$ follows at once. Since α_2 is an involution,

$$\theta(\sigma_2) = \theta(\alpha_1^{-1}\alpha_2) = \alpha_1\alpha_2 = \sigma_1^2\sigma_2.$$

For $i \geq 3$ both α_{i-1} and α_i are involutions, so they are fixed by θ, and their product σ_i is fixed. Conversely the displayed formulas give $\theta(\alpha_1) = \alpha_1^{-1}$ and $\theta(\alpha_2) = \sigma_1^{-1}\sigma_1^2\sigma_2 = \alpha_2$. Multiplying by the fixed later generators gives $\theta(\alpha_i) = \alpha_i = \alpha_i^{-1}$ for every $i \geq 2$. Applying θ twice fixes the generating set (α_i). □

Thus the two reflection tests agree whenever the appropriate string-group reconstruction theorem applies. We retain that reconstruction theorem, including its intersection hypotheses, as an attributed input. A presentation with the displayed relations alone is not a reconstruction proof. Also θ may be the identity; in that case a geometric reflection interchanging two chamber orbits is still a separate involution, as in the accepted residue theorem.

2 Automorphisms and adjacency operations

For a polyhedron let r_i send a flag to its unique i-adjacent flag. The symbols r_i in this section denote permutations of flags. They need not be automorphisms of the polyhedron. They satisfy $r_i^2 = 1$ and $r_0 r_2 = r_2 r_0$. A word is applied in its written order; write Φ^w for its result on a flag Φ. Automorphisms commute with all these operations because adjacency is intrinsic.

Lemma 2. *Suppose the automorphism group acts freely and transitively on a set of flags $\mathcal{O}$, and a word w preserves $\mathcal{O}$. Let g_w be the unique automorphism with $g_w\Phi = \Phi^w$. Then $\Phi^{w^k} = g_w^k\Phi$. The least positive return time of this flag walk is the order of g_w, with both interpreted as infinite if there is no return.*

Proof. An automorphism fixing one flag fixes its adjacent flags and hence all flags by connectedness; thus the freeness hypothesis holds in every application here. Commutation gives $(g_w^k\Phi)^w = g_w^k(\Phi^w) = g_w^{k+1}\Phi$, proving the identity by induction. The final assertion is freeness. □

In a regular polyhedron $\mathcal{O}$ is the full flag set. In a chiral polyhedron there are two such orbits and every adjacency interchanges them. A word of even length preserves each orbit, and a word of odd length interchanges them. Accordingly the lemma applies to even words in the chiral case.

3 Holes and zigzags

A flag records a vertex, an incident edge and one incident face. Crossing the edge is the operation r_0. At the new vertex, the operation r_1 takes the next edge in the selected face, and r_2 changes the selected face while keeping the vertex and edge. Repeating these two local operations goes around the vertex figure.

Fix $1 \le j \le \lfloor q/2 \rfloor$ for a polyhedron of type $\{p, q\}$. The following words each advance by one edge of the indicated walk:

$$h_j = r_0 r_1 (r_2 r_1)^{j-1}, \qquad z_j = r_0 (r_1 r_2)^j.$$

Indeed after r_0, the first r_1 selects the first outgoing edge; each additional $r_2 r_1$ selects the next one. This gives the jth outgoing edge. In h_j the selected face remains on the same side of the walk, and the successive turns have the same sense. The final r_2 in $z_j = h_j r_2$ chooses the opposite incident face, so the next turn has the opposite sense. These are respectively the j-hole and j-zigzag rules. Their lengths are their complete traversal periods, including the selected side of the traversal, rather than the first return to a previously visited vertex.

For a regular polyhedron, let ρ_i be the distinguished automorphism with $\rho_i\Phi = \Phi^{r_i}$. Commutation with adjacency shows inductively that a word in the ρ_i has the same effect on Φ as the written word in the r_i. Lemma 2 therefore gives

$$\ell(h_j) = \operatorname{ord}\bigl(\rho_0\rho_1(\rho_2\rho_1)^{j-1}\bigr), \qquad \ell(z_j) = \operatorname{ord}\bigl(\rho_0(\rho_1\rho_2)^j\bigr).$$

For a chiral polyhedron choose the distinguished rotations by $\sigma_1\Phi = \Phi^{r_0 r_1}$ and $\sigma_2\Phi = \Phi^{r_1 r_2}$. These are consecutive rotations with the chosen base orientation. Write $s_1 = r_0 r_1$, $s_2 = r_1 r_2$ as formal words in involutions. Direct cancellation gives

$$h_j = s_1 s_2^{1-j}, \qquad r_0 s_2^j r_0 = s_1 s_2^{-j} s_1^{-1}, \qquad z_j^2 = s_1 s_2^{-j} s_1^{-1} s_2^j.$$

For $j = 1$, direct cancellation gives

$$(r_0 r_1)(r_2 r_1)(r_1 r_0) = r_0 r_1 r_2 r_0 = r_0 s_2 r_0.$$

Taking the jth power proves the middle identity. The first and last words are even. Applying the lemma on the base flag orbit proves

$$\ell(h_j) = \operatorname{ord}(\sigma_1 \sigma_2^{1-j}),$$
$$\ell(z_j) = 2\operatorname{ord}(\sigma_1 \sigma_2^{-j} \sigma_1^{-1} \sigma_2^j).$$

The factor two is necessary: one zigzag step exchanges the two flag orbits, so a return cannot have odd length. The alternative word $\sigma_2^j \sigma_1 \sigma_2^{-j} \sigma_1^{-1}$ is conjugate to the word displayed above, by σ_2^j, and has the same order.

These computations identify the exact flag periods. They neither say that the period data determine the polyhedron nor prove finiteness of a universal quotient with prescribed periods. Those are the separate questions posed in the survey.

References

[1] E. Schulte and A. Ivić Weiss, *Problems on polytopes, their groups, and realizations*, Period. Math. Hungar. 53 (2006), 231–255; fixed arXiv math/0608397v1, Sections 2–3. We use the survey's definitions and its explicit attribution of the reconstruction theorem to their earlier work on chiral polytopes.